Lab 5

Pathogenwatch: Many Genomes at Once

Day 2 · 11:30–12:30 · 60 min
Pairs

LO5: use a free web platform to type, screen and visualise a collection of genomes

Open your handout

Lab 5 handout

`Lab5_pathogenwatch.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

The scenario

- The IPC nurse at **District Hospital A** is worried about more **MRSA SSIs on Surgical Ward B**
- The lab stored the isolates and sent them for sequencing
- You have **genomes from all MRSA isolates of the past months**

KEY MESSAGE

Get a quick overview of all the genomes at once.

The genomes are real public data. The hospital, wards and dates are fictional.

Plan for this lab

Part	What	Time
A	Upload: typing and AMR	30 min
B	Tree + timeline + metadata	25 min
	Summary	5 min

Files from the instructor

File	Contains
`assemblies/`	One FASTA per isolate
`lab6_metadata.csv`	Patient, ward, procedure and date information
`koser_snp_matrix.tsv`	SNP distances (used in Lab 6)

- **GitHub:** github.com/aryakaul/ughe-ssi-genomics-course
- Green **Code** button → **Download ZIP** → unzip → `labs/data/`

Upload and explore

1. Log in to `pathogen.watch` → **Upload**
2. Drag in the FASTA assemblies **and** `lab6_metadata.csv` **together**
3. For **at least 6** genomes, record:

Genome	Species	ST	Predicted resistances	QC pass?

Does the organism agree with the lab? Check the assembly QC metrics (length, contigs, N50).

Make a collection and view the tree

1. Select all your genomes → **Create collection** → name it `UGHE Lab 5`
2. Open the collection → explore the **tree** view
3. Explore the **AMR** view: resistance profiles side by side

CHECK YOURSELF

1. How many different STs are in the collection?
2. Do all isolates have the **same** AMR profile? What does that suggest, and what does it **not** prove?
3. In the tree, can you see a group of isolates on very **short** branches?

Find your metadata

1. Open the `UGHE Lab 5` collection
2. Look at the **data tables** at the bottom of the page
3. Find the columns `ward`, `theatre` and `hospital`

> No metadata columns? Raise your hand.

Explore: tree + time + place

- **Colour** the tree: click the `ward` column header, then `theatre`, then `hospital`
- Open the **Timeline**: when did each infection happen?
- **Click a tip**: find it in the table and timeline

CHECK YOURSELF

1. Is there a group that is closely related **and** close in time **and** in the same place?
2. Is any isolate from the **same ward at the same time not** in that group?
3. Is any isolate in the group from a **different ward**? What could link it?

PART B3

Save, share, summarise

Copy the link to your collection from the address bar. **You'll use it in Lab 6.**

Pathogenwatch gives you	Input
Species, ST, AMR, QC, clustering, global context	FASTA assemblies
A visual story: tree + time + place + metadata	Assemblies + a table (CSV)

Free and used by public-health labs. **You can use it on Monday with your own data.**

Pathogenwatch: Centre for Genomic Pathogen Surveillance.